

NAITIK GUPTA

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SUMMARY

AI engineer who builds AI systems end to end on AWS (LLM agents, RAG and text-to-SQL, vision-language and computer-vision pipelines, recommender serving and load testing) and the evaluation harnesses that prove they work. Also the technical voice on pre-sales calls, turning client problems into AWS architectures, cost estimates and SOWs. B.S. Computer Engineering, University of Maryland.

EXPERIENCE

Minfy Technologies (AWS Consulting Partner)

Tysons, VA

Solutions Architect (Pre-Sales)

Sep. 2026 – Present

- Technical voice on AWS pre-sales calls alongside the sales team: run discovery, present Minfy's AI/ML and migration work, and turn client problems into proposed AWS architectures, cost estimates and SOW scope.
- Pitched Minfy's GenAI and AWS modernization work at Amazon HQ2, using my restaurant-platform project as the reference; led to follow-up prospect calls.
- Wrote the SOW for an AI marketing-studio POC (Bedrock Claude, Nova Canvas, deterministic HTML renderer, social publishing connectors, MCP server, human approval gate): 10 measurable objectives, 4 gated phases, 15 deliverables.
- Built multi-tier AWS Pricing Calculator estimates and POC budgets for IoT, GenAI-marketing and document-AI proposals (e.g. 20-service IoT POC: \$210 to \$1,244/month across 100 to 5,000 devices).
- Wrote an AWS migration assessment for a 10-service startup stack: spend scenarios, per-service AWS substitutes, hidden costs and funding fit (MAP vs Activate).

AI Engineer

Jan. 2026 – Present

Grubhub – Personalized Topics Ranking (TensorFlow multi-task ranker)

- Owned preprod load and latency testing of the personalized ranker (SageMaker behind Grubhub's model platform; Spinnaker, Jenkins/Gatling, Datadog, CloudWatch): **1.1M requests** in 3 hours, **166 req/s peak (1.7× production's peak)**, **3.9 ms p95, 0.0002% errors**.
- Traced an apparent 30 req/s ceiling to test-harness start-up congestion and the auth service, not the model, by reading the Gatling source and per-stage Datadog metrics; the corrected method raised measured throughput **5×**.
- Proved with a 5-job follow-up that the remaining ~160 req/s cap lived in the test harness; root-caused a 100%-failure wave to a stale platform version and shipped a pre-flight check.
- Fixed a partition-read bug in the Spark-on-EMR training pipeline that silently zero-filled every order-history feature (**0.36M rows read instead of 167.9M**); fixed three Spark SQL bugs and enabled AQE on a stage stuck 90+ minutes.
- Got the production retraining ETL (Azkaban, Jenkins, EMR) running in dev after root-causing a bash brace-expansion bug corrupting JSON job config; modeled per-run cluster cost.
- Clickstream EDA over 5.7M impressions/day (U-shaped CTR-by-rank position bias); built a multi-environment Redash CLI and a CLI handbook that sets up the full toolchain in ~20 minutes.

Omaha Productions – Broadcast Ratings VLM Pipeline (ManningCast)

- Built in three days a pipeline that segments a 3.5-hour broadcast (ffmpeg black-frame detection), adds YOLO11 evidence, tags guests, commercials and scorebugs with Claude on Bedrock, and joins every minute to Nielsen ratings.
- Tagged 241 windows with zero refusals and produced a 185/185-minute ratings-to-video alignment proof that resolved a client concern about a telecast offset; resumable CLI and interactive HTML report.

Samsara – Dashcam Safety Analysis (VLM + Computer Vision)

- Built a cost-aware CV preprocessing stack for VLM video analysis: frame-rate reduction with a Gradio UI, YOLOv8 + ByteTrack tracking, approach/near-miss heuristics, bounding-box overlays and structured JSON context.
- Built a tri-state (yes/no/unknown) rule-based extractor for a 45-label safety taxonomy; showed stock detection answers ~12% of labels (~19% with raw-video signals), shaping model selection.
- Designed a 3-track × 4-VLM evaluation on 97 labeled clips with a YOLO ground-truth audit layer; wrote the scorer (precision/recall/F1, confusion matrices, cross-run leaderboard) and the client ground-truth framework report.

AI Evaluation Test Bench

- Primary author (21/22 commits) of a black-box evaluation bench for any RAG, multi-turn or text-to-SQL endpoint: ingests folders, ZIPs or S3 into ChromaDB baselines and auto-generates questions and follow-ups.
- Three-layer scoring (cosine similarity and number matching; Top-K, number and spaCy grounding for hallucination risk; LLM-as-judge on five rubrics) plus an independent LLM SQL oracle; FastAPI UI, CLI and Docker.

Restaurant Reservation Marketplace – POS Integration and Chat-to-BI

- Built Square and Clover POS integrations in TypeScript/Express: per-restaurant OAuth 2.0, address-scored location matching, idempotent order sync, check-status sync and self-healing cross-vendor customer dedup.
- Built an 8-node Chat-to-BI assistant on Bedrock: LLM intent router, deterministic SQL report nodes, an AST-validated tenant-scoped LLM SQL node, auto-generated Plotly charts and a React chat UI with PDF export.

Zillion Technologies

Ashburn, VA

AI Engineer

Oct. 2025 – Jan. 2026

- Built an agentic voice agent for Sysco in LangGraph and Google ADK, connecting Salesforce and Talkdesk (websockets) so callers query live orders in natural language mid-call; deployed on Vertex AI.
- Replaced a hosted GPT call with Phi-3 Mini on a 6 GB GPU by moving from FP16 offload to 4-bit NF4 (~3 GB VRAM); tuned decoding for determinism and shipped a side-by-side GPT comparison, keeping IAM data off external APIs.

Software AI Engineer Intern

Jun. 2025 – Oct. 2025

- Built the admin panel for Zecured, an AI-assisted identity-governance product: Okta, Saviynt and SailPoint onboarding/offboarding with RBAC and least privilege.
- Clustered hundreds of roles by entitlements and descriptions (regex, fuzzy and semantic matching), added AI-suggested access levels and role merges, and a two-stage PO/Manager approval flow.

Software Engineering Intern

Jun. 2024 – Aug. 2024

- Delivered a client-facing SharePoint search system: optimized SharePoint API crawl, OCR and speech-to-text extraction, MiniLM embeddings in ChromaDB, and a FastAPI + OpenAI RAG chatbot returning answers with file locations.
- Added delta updating so only modified files and folders are re-indexed; managed delivery in Agile/JIRA with the project manager and stakeholders.

Engineering IT (TerrapinWorks), University of Maryland

College Park, MD

Lead Software Developer

May 2024 – May 2025

- Led a 10-person Scrum team shipping 4 production features for engineering labs (sign-in reasons with auto sign-out, post-visit surveys, admin rule controls, redesigned machine catalog).
- Owned hiring end to end (screening, technical interviews, decisions), code review and GitLab CI/CD releases; turned feature requests into frontend/backend tickets and paired juniors with seniors.
- Built the onboarding curriculum with Canvas videos: Git, Linux/WSL, Docker, local setup, branching and merge workflows.

Software Development Engineer

Aug. 2022 – May 2024

- Built Pinpoint (React, Flask/FastAPI, MySQL, Docker): magnetic-stripe swipe login with usage logging and Canvas Badges API training gates for tool access across UMD labs, used by 100+ students and staff.
- Redesigning machine categorization across backend and UI; worked on an order-management system for 3D-printed parts (OutSystems); wrote internal docs and handled support tickets.

Undergraduate Researcher, Prof. Romel Gomez, UMD ECE

College Park, MD

IoT Communication over GSM

May 2023 – 2024

- Built SMS-based appliance control over GSM (SIM7600A-H, Arduino, C) with security and range as goals; designed a 3D-printed enclosure and soldered all hardware.

PROJECTS

The Quintacle – Local-First Personal AI OS | Python, FastAPI, Next.js 16, React 19, SQLite, Claude, OpenAI 2024 – 2026

- “Deterministic core, LLM at the edges” knowledge system: 7.7k Python + 9.5k TypeScript LOC, 101 REST routes, 12 domain modules, 21 pages, 138 tests.
- Regex routing plus Claude structured outputs turn text, photos, PDFs, DOCX and video into typed entries, each human-approved before write; markdown is the source of truth with a rebuildable SQLite index.
- Code-computed knowledge graph (d3-force), EXIF date/GPS photo filing, multi-provider chat with web search and prompt caching, nutrition tracking against 20 micronutrient RDAs. Rebuilt from a 2024 voice-driven LLM + RAG v1.

NaviGatr – Edge-AI Wearable for the Visually Impaired | Raspberry Pi 5, Coral TPU, TFLite, AWS EC2 Jan. – May 2025

- Capstone headset fusing object detection (SSD MobileNet on Coral TPU), monocular depth (Apple Depth Pro on EC2) and emotion recognition into clock-face audio (“person, 2 m, 10 o’clock, happy”).
- Trained the EfficientNet emotion model on FER2013; designed and 3D-printed the headset (Pi, TPU, camera, cooling, power, display); configured Linux, TPU runtimes and multithreaded inference; recovered a bricked Pi via EEPROM.

LLM Evaluation Toolkit | Python, SciPy, Typer, pytest, Anthropic/OpenAI/Ollama

Oct. 2026

- Five offline-reproducible evaluation tools (292 tests): failure-taxonomy classifier with human review queue, LLM-judge reliability study (position/verbosity bias, kappa), regression gate with bootstrap CIs and McNemar tests, inter-annotator agreement benchmark, RAG quality evaluator.

PeopleLens – Visual Contact Recognition | PyTorch, FaceNet, Roboflow, FastAPI, SQLite

Sep. 2026

- Consent-based face re-identification: Roboflow/MTCNN detection, FaceNet embeddings, cosine matching with thresholds and unknown-person rejection; FastAPI endpoints with local-only storage.

Offline LLM Agent | Ollama, Qwen3, FastAPI, RTX 5070 Ti

Sep. 2026

- Fully local chat agent with persistent memory and a bounded tool-calling web-research loop (search, fetch, cite) running Qwen3 14B on a 12 GB laptop GPU.

naitikg.us – Portfolio with RAG Chatbot | *Next.js, FastAPI, ChromaDB, Vercel, Render* 2025 – Present

- Markdown-driven site with an embedded RAG chatbot over its own content, an in-browser 3D STL viewer and doc viewer; frontend on Vercel, chatbot API on Render, auto-deployed from GitHub.

Self-Balancing Robot | *ESP32, MPU6050, C++, PID, Fusion 360* Apr. – May 2025

- Two-wheel inverted-pendulum robot: IMU calibration, PWM motor control, dead zone and tilt cut-offs; tuned the PD loop from logged telemetry; designed the 3D-printed chassis.

Autonomous Crash-Site Rover (ENES100) | *Arduino, ultrasonic/IR sensors, H-bridges* Aug. – Dec. 2021

- Conceived the 4-motor tank-tread drive that shaped the rover; led electronics, introducing hot-swappable controller sockets and a second H-bridge; detected the anomaly and deployed the ArUco marker.

Hardware and Home Builds | *ESP32, Raspberry Pi, Alexa API, Arch Linux, Bambu A1* 2022 – 2024

- Alexa-synced automated window blinds (Pi + Flask, then ESP32) with 3D-printed gears; Arch Linux + Apache home web server; electromagnetic pulse generator; 3D-printed desk and storage systems; Android Auto retrofit and DIY car maintenance.

AWARDS

2nd Place, AWS × Anthropic Energy Hackathon (Claude Code + Bedrock AgentCore) | Houston, TX Sep. 2026

- Built the detection and leak-localization tools for a pipeline leak-detection agent (Strands + Claude Sonnet 4.5) and deployed it to Bedrock AgentCore: 5/5 leaks caught, 15/15 false alarms rejected, ~5-mile average localization on a 200-mile line.
- Coordinated a three-engineer team through Claude Code with a shared CLAUDE.md git-sync protocol, going from scaffold to a deployed, tested agent in one build day.

EDUCATION

University of Maryland, College Park College Park, MD

Bachelor of Science in Computer Engineering Aug. 2021 – May 2025

- Coursework: Machine Learning, Artificial Intelligence, ML Design Capstone, Operating Systems, Digital Computer Design, Computer Organization, Microprocessors, Control Systems, Algorithms, Signals and Systems, Programming Languages.
- Course: Qubit by Qubit Introduction to Quantum Computing (2022).

LEADERSHIP & ACTIVITIES

President, Cloud Computing Club, UMD Apr. 2023 – May 2025

- Led 12 VPs running AWS-focused tutorials and certification prep for 300+ members; planned contract-based cloud projects with students.

President, Resident Housing Association, South Campus Commons Aug. 2022 – May 2023

- Led weekly senate meetings (team of 12) and planned 4+ events a semester for 2,000+ residents.

Student Ambassador, A. James Clark School of Engineering Aug. 2022 – 2023

- Campus tours, open houses, Q&A panels and recruitment events for admitted students and families; ClarkLead admit events.

Activities: Pi Lambda Phi, UMD Tenor & Bass Choir, UMD Club Sailing, Enrage runway modeling (2022 – 2023)

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, SQL (Presto/Trino, Spark SQL, MySQL, SQLite), Bash, C/C++, Java

LLMs & Agents: Claude API (tool use, structured outputs, vision, caching), Amazon Bedrock, Bedrock AgentCore, Strands Agents, LangGraph, Google ADK, OpenAI, RAG, ChromaDB, SentenceTransformers, Ollama, Hugging Face, 4/8-bit quantization, Claude Code

Evaluation: LLM-as-judge, golden-set regression testing, RAG metrics (recall@k, MRR, nDCG, faithfulness), precision/recall/F1, load and latency testing (Gatling)

ML & Vision: PyTorch, TensorFlow/Keras, TFLite, YOLO (v8, YOLO11) + ByteTrack, FaceNet, EfficientNet, Depth Pro, OpenCV, spaCy, scikit-learn, recommender systems

Cloud & Data: AWS (Bedrock, SageMaker, EC2, EMR, S3, Cognito, CloudWatch, Pricing Calculator), Vertex AI, Spark, Azkaban, Jenkins, Spinnaker, Datadog, Redash

Engineering: FastAPI, Flask, Express, React, Next.js, Docker, Linux (Arch, Ubuntu, WSL2), Git, GitLab CI/CD, REST, OAuth 2.0, Salesforce/Talkdesk/Okta/Square/Clover APIs

Hardware: Raspberry Pi, Coral TPU, ESP32, Arduino, IMUs, GSM modules, PID control, Fusion 360, 3D printing, soldering